

Inequalities in Real-World Contexts

MA.8.AR.2.2

I can write an inequality from a situation, solve it, and say what the answer means.

The last lesson of the unit.

Everything from Lessons 6 to 9, in one question at a time.

Warm-Up

On your own. Four minutes.

Same context, different question

A ride costs \$4 to start plus \$2.50 a mile.

- 1.** What does a 6-mile ride cost?
- 2.** A ride cost exactly \$24. How far was it?
- 3.** You have at most \$25. How far can you go?

Notes

I. The phrase attached to the limit gives the symbol

Words	Symbol	Example
at most, no more than, maximum	$\leq$	at most \$25 $\rightarrow$ total $\leq$ 25
at least, no less than, minimum	$\geq$	at least 170 shirts $\rightarrow$ shirts $\geq$ 170
fewer than, less than, under	$<$	fewer than 12 people $\rightarrow$ people $<$ 12
more than, greater than, over	$>$	more than 3 hours $\rightarrow$ hours $>$ 3

Notes

II. Build it the same way as an equation, then add the symbol

- A. **Name the unknown in a sentence, exactly as in Lessons 6 and 7.**
- B. Find the rate and the one-time amount, exactly as before.
- C. **Find the limit, and read the phrase attached to it. That phrase gives the symbol.**
- D. Total on the left, limit on the right.

Notes

III. Solving does not change at all

- A. **The symbol behaves exactly as it did in Lessons 8 and 9.**
- B. Add and subtract freely. Multiply and divide by a positive freely.
- C. **Reverse only when you multiply or divide both sides by a negative.**

Notes

IV. The answer is a sentence, and it may need rounding

- A. **Solve, then ask what the number counts.**
- B. A maximum rounds DOWN. A minimum rounds UP. The decimal never decides.
- C. **Justify with arithmetic on both sides — the value that works, and the next one that does not.**

Example 1

A budget.

A ride costs \$4 to start plus \$2.50 a mile. You have at most \$25.

Write the inequality. Do not solve it.

Define your letter, and say which phrase gave you the symbol.

Example 1

Solve it. Anything flip?

$$4 + 2.5m \leq 25$$

$$2.5m \leq 21$$

subtracted 4 — no flip

$$m \leq 8.4$$

divided by POSITIVE 2.5 — no flip

Example 1

Now answer the question.

$m \leq 8.4$ is the solution set.

8 miles costs $4 + 2.5(8) = \$24$ within budget ✓

9 miles costs $4 + 2.5(9) = \$26.50$ over budget ✗

You can go 8 whole miles.

Example 2

Now a minimum.

A store must keep at least 170 shirts on display. It has 50, and each box holds 24.

How many boxes must be opened?

Example 2

Solve it.

$$50 + 24b \geq 170$$

$$24b \geq 120$$

$$b \geq 5$$

Example 2

And if the store had 60 shirts?

$$60 + 24b \geq 170$$

$$24b \geq 110$$

$$b \geq 4.58\dots$$

4 boxes gives $60 + 96 = 156$ shirts not enough ✗

5 boxes gives $60 + 120 = 180$ shirts enough ✓

Example 3

Two answers, both 7.4.

At most 7.4 cases of juice fit the budget.	At least 7.4 crates are needed.
How many cases do you buy?	How many crates do you order?

Same number. Same rounding?

Example 3

No.

7 cases	8 crates
8 would go over the budget	7 would not be enough
a maximum rounds DOWN	a minimum rounds UP

Example 3

The symbol tells you which way.

$\leq$ a maximum	$\geq$ a minimum
round DOWN	round UP
going over breaks the limit	falling short breaks the limit

Whiteboards • Question 1 of 9

Boards up on three.

She has at most \$25 to spend.

Answer it.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 1 of 9

Answer.

She has at most \$25 to spend.

“at most” includes the 25

Answer: ≤ 25

Whiteboards • Question 2 of 9

Boards up on three.

She has at least \$25 to spend.

Answer it.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 2 of 9

Answer.

She has at least \$25 to spend.

“at least” also includes the 25

Answer: ≥ 25

Whiteboards • Question 3 of 9

Boards up on three.

She has more than \$25 to spend.

Answer it.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 3 of 9

Answer.

She has more than \$25 to spend.

“more than” EXCLUDES the 25

Answer: > 25

Whiteboards • Question 4 of 9

Boards up on three.

A ride costs \$4 plus \$2.50 a mile. At most \$25.
Write it.

Answer it.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 4 of 9

Answer.

A ride costs \$4 plus \$2.50 a mile. At most \$25.
Write it.

the phrase from question 1 gives $\leq$

Answer: $4 + 2.5m \leq 25$

Whiteboards • Question 5 of 9

Boards up on three.

Solve $4 + 2.5m \leq 25$

Answer it.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 5 of 9

Answer.

Solve $4 + 2.5m \leq 25$

$2.5m \leq 21$, divided by POSITIVE 2.5

Answer: $m \leq 8.4$

Whiteboards • Question 6 of 9

Boards up on three.

A ride costs \$4 plus \$2.50 a mile. At most \$25, so $m \leq 8.4$.
How many whole miles?

Answer it.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 6 of 9

Answer.

A ride costs \$4 plus \$2.50 a mile. At most \$25, so $m \leq 8.4$.
How many whole miles?

8 miles costs \$24; 9 miles costs \$26.50

Answer: 8 miles

Whiteboards • Question 7 of 9

Boards up on three.

A club needs to raise at least \$200. Each car wash brings in \$35. How many washes?

Solve it, then answer the question.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 7 of 9

Answer.

A club needs to raise at least \$200. Each car wash brings in \$35. How many washes?

$$35w \geq 200, \text{ so } w \geq 5.71\dots$$

Answer: 6 washes

Whiteboards • Question 8 of 9

Boards up on three.

$$-\frac{2}{3}(-3 - 12a) \leq -6$$

Answer it.

A symbol, an inequality, a solution set, or a number of things — read which.

Whiteboards • Question 8 of 9

Answer.

$$-\frac{2}{3}(-3 - 12a) \leq -6$$

$\times -3/2$, REVERSE; then $\div -12$, REVERSE again

Answer: $a \leq -1$ two flips

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Quinn solved $2.5 \geq (c - 3) \div 4$ by adding 3 to both sides and got $22 \geq c$. What went wrong?

Write your answer in sentences.

This one is written work. Say what she did right as well.

Whiteboards • Question 9 of 9

Answer.

Quinn solved $2.5 \geq (c - 3) \div 4$ by adding 3 to both sides and got $22 \geq c$. What went wrong?

multiply by 4 first: $10 \geq c - 3$

Answer: $c \leq 13$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.